

Lecture#09

MATHEMATICAL EXPECTATION

If X is a discrete random variable with the probability function $f(x)$, then the expected value or the mathematical expectation of X , $E(X)$ is defined as,

$$E(X) = \sum_x xf(x)$$

If X is continuous having a density function $f(x)$, then

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

*If C is a constant, $E(C) = C$

* $E(C \times g(x)) = E(C) \times E(g(x)) = C \times E(g(x))$

* $[EW_1(X) + W_2(X) + \dots + W_k(X)] = E[W_1(X)] + E[W_2(X)] + \dots E[W_k(X)]$

$$E[W(X)] = \left\{ \begin{array}{l} \sum_x W(x)f(x), \text{ if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} W(x)f(x)dx, \text{ if } X \text{ is continuous} \end{array} \right\}$$

*The Variance of a random variable x is

$$V(X) = E[(X - \mu)^2]$$

$$= E(X^2) - \mu^2,$$

where μ is the expected value.

Example

X	-3	-2	0	1	2
$P(X = x)$ $= f(x)$	0.10	0.30	0.15	0.40	0.05

Find $E(X)$ and $V(X)$.

Solution:

$$\mu = E(X) = \sum_x xf(x)$$

$$= (-3 \times 0.1) + (-2 \times 0.30) + (0 \times 0.15) + (1 \times 0.40) + (2 \times 0.05)$$

$$= -0.4$$

$$V(X) = E(X - \mu)^2 = \sum_{-3}^2 (X - \mu)^2 xf(x)$$

$$= (-3 + 0.4)^2 \times 0.1 + (-2 + 0.4)^2 \times 0.30 + \dots + (2 + 0.4)^2 \times .05$$

$$= 2.54$$

$$\text{Or, } E(X^2) = \sum_x x^2 f(x)$$

$$= (-3)^2 \times .10 + (-2)^2 \times .30 + \dots + 2^2 \times .05$$

$$= 2.7$$

$$\therefore V(X) = E(X^2) - \mu^2 = 2.7 - (-.4)^2 = 2.54$$

■ Standard deviation (σ) is the square root of the variance, i.e.

$$\sigma = \sqrt{V(X)} = \sqrt{2.54} = 1.59 .$$

Example: A life insurance company in Bangladesh offers to sell a TK.25000 one year term life insurance policy to a 25 year old man for a premium of TK.2500. According to Bangladesh life table, the probability of serving one year for a 25 year old man is 0.97. What is the company's expected gain in the long run?

Solution:

The gain X is a random variable that may take on the values 2500, if the man survives or 2500-25000=-TK.22500 if he dies. Consequently, the probability of X is

$$\begin{array}{lll} X & : & 2500 \qquad \qquad -22500 \\ f(x) & : & 0.97 \qquad \qquad \qquad 0.03 \end{array}$$

$$\text{So, } E(X) = (2500 \times .97) + (-22500 \times .03) = 1750$$

Thus, the ultimate gain of the company is 1750.

Example: Find the mean (expected value) and the variation of the following function

$$f(x) = 2(1 - x), 0 < x < 1$$

Solution:

$$E(X) = \int_0^1 x(2x - 1) dx = 2 \left[\int_0^1 x dx - \int_0^1 x^2 dx \right] = \frac{1}{3}$$

$$E(X^2) = \int_0^1 x^2 \cdot 2(1 - x) dx = \frac{1}{6}$$

$$\therefore V(X) = \sigma^2 = E(X^2) - [E(X)]^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}$$

Example: Given the following discrete distribution

$$f(x) = \frac{e^{-m} m^x}{x!}, \quad x = 0, 1, 2, \dots \infty$$

Find $E(X)$.

Solution:

By definition

$$\begin{aligned} E(X) &= \sum_{x=0}^{\infty} xf(x) = \sum_{x=0}^{\infty} \frac{xe^{-m}m^x}{x!} = e^{-m} \sum_{x=0}^{\infty} \frac{xm^x}{x!} \\ &= e^{-m}m \left(1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots \right) = e^{-m} m = m. \end{aligned}$$

Example: A lot of 7 markers is supplied by a quality inspector; the lot contains 4 good markers and 3 defective markers. A sample of 3 is taken by the inspector. Find the expected value of the number of good markers in this sample.

Solution:

Let X represent the number of good markers in the sample. It can be shown that the probability distribution of X is

$$f(x) = \frac{{}^4C_x {}^3C_{3-x}}{{}^7C_3}, \quad x = 0, 1, 2, 3.$$

Calculation shows that the probability distribution of X is as shown in the accompanying table:

Values of X	0	1	2	3
f(x)	$\frac{1}{35}$	$\frac{12}{35}$	$\frac{18}{35}$	$\frac{4}{35}$

$$\text{Therefore, } E(X) = (0) \left(\frac{1}{35} \right) + (1) \left(\frac{12}{35} \right) + (2) \left(\frac{18}{35} \right) + (3) \left(\frac{4}{35} \right) = 1.7.$$

Thus, if a sample of 3 markers is selected at random over and over again from a lot of 4 good markers and 3 defective markers, it would contain, on average 1.7 good markers.

Example: In a coin tossing program, a man is promised to receive TK. 5 if he gets all heads or all tails when three coins are tossed and he pays off (loses) TK. 3 if he either one or two heads appear. How much is he expected to gain in the long run?

Solution: The random variable here is the amount the man can win. If X is the random variable, then X will be on a value 5 when the coins show all heads and -3, otherwise. The table below shows the outcomes of the experiment, values of X and the associated probabilities:

Outcome:	HHH	HTT	HTH	HHT	THH	THT	TTH	TTT
X	5	-3	-3	-3	-3	-3	-3	5
$f(x)$	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8

It appears from the above table that the variable X assumes values -3 and 5 with probabilities 6/8 and 2/8 respectively. Since the value -3 occurs 6 times and 5 occurs 2 times the expected value of X is

$$E(X) = \sum xf(x) = -3\left(\frac{6}{8}\right) + 5\left(\frac{2}{8}\right) = -1.$$

Thus, the man is expected to lose TK. 1 in the long run.

Let us now examine what happens if the man receives TK. 5 for all heads or all tails, Tk. “0” for 2 heads and pays off Tk. 3 for 1 head. The random variable X will now assume the values, 5, 0 and -3 with associated probabilities 2/8, 3/8 and 3/8 respectively. The expected value in this case will be

$$E(X) = \sum xf(x) = 5\left(\frac{2}{8}\right) + 0\left(\frac{3}{8}\right) + (-3)\left(\frac{3}{8}\right) = \frac{1}{8} = 0.125.$$

This shows that the man will be marginally gainer winning only 12.5 paise.

EXPECTED VALUE OF A FUNCTION OF TWO RANDOM VARIABLE

The notion of mathematical expectation can be expended of two or more random variables. We will deal here with the case of two variables, which can be

analogously being extended for 3 or more variables. Let X and Y be two random variables with joint probability distribution $f(x, y)$. the expected value of the function $w(X, Y)$ is defined as

$$E[w(X, Y)] = \sum_x \sum_y w(x, y)f(x, y), \text{ if } X \text{ and } Y \text{ are discrete}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} w(x, y)f(x, y) dx dy, \text{ if } X \text{ and } Y \text{ are continuous}$$

Further if $w(X, Y)$ is a function of the random variable X and Y , and c is a constant, then

$$E[cw(X, Y)] = cE[w(X, Y)] = c \sum_y \sum_x w(x, y)f(x, y)$$

And, also, if X and Y are two random variable and $w_1(X, Y), w_2(X, Y)$ are the functions of X and Y , then

$$E[w_1(X, Y) + w_2(X, Y)] = E[w_1(X, Y)] + E[w_2(X, Y)].$$

Theorem 1: The expected value of the sum of two random variables X and Y is the sum of the expected values of the random variables. Symbolically,

$$E(X + Y) = E(X) + E(Y).$$

Corollary 1: If $X \geq Y$, then $E(X) \geq E(Y)$

Corollary 2: If a and b are two constants, then $E(aX + bY) = aE(X) + bE(Y)$

Corollary 3: If $a_1, a_2, \dots, a_k$ are k constants, then for k RVs $X_1, X_2 \dots X_k$

$$E\left[\sum_{i=1}^k a_i X_i\right] = \sum_{i=1}^k a_i E(X_i).$$

Theorem 2: The expected value of the two random variables X and Y is equal to the product

of their expected values, only when the variables are independent, i. e

$$E(XY) = E(X)E(Y)$$

Or, in other words,

The expected value of the product of two random variable is equal to the product of their expectations.

Example: Given the following density function

$$f(x, y) = 2(x + y - 2xy), 0 < x < 1, 0 < y < 1$$
$$= 0, \text{ elsewhere}$$

- (a) Find $E(X)$, $E(Y)$, $E(X + Y)$ and $E(XY)$
- (b) Also verify whether $E(X + Y) = E(X) + E(Y)$
- (c) Are X and Y independent?

Solution:

The marginal density of X is

$$g(x) = 2 \int_0^1 (x + y - 2xy) dy = 2 \left(x + \frac{1}{2} - x \right) = 1$$

Hence,

$$E(X) = \int_0^1 xg(x)dx = \int_0^1 xdx = \frac{1}{2}$$

Similarly,

$$h(y) = 2 \int_0^1 (x + y - 2xy) dx = 2 \left(\frac{1}{2} + y - y \right) = 1$$

And

$$E(Y) = \int_0^1 yh(y) dy = \int_0^1 y dy = \frac{1}{2}$$

$$\begin{aligned} E(X + Y) &= 2 \int_0^1 \int_0^1 (x + y)(x + y - 2xy) dx dy \\ &= \int_0^1 \int_0^1 (x^2 + y^2 + 2xy - 2x^2y - 2xy^2) dx dy \end{aligned}$$

$$= 2 \int_0^1 \left(x^2 + \frac{1}{3} + x - x^2 - \frac{2x}{3} \right) dx$$

$$= 2 \int_0^1 \left(\frac{1}{3} + \frac{x}{3} \right) dx = 1$$

Since $E(X) + E(Y) = \frac{1}{2} + \frac{1}{2} = 1$, we establish that $E(X + Y) = E(X) + E(Y)$

$$E(XY) = \int_0^1 \int_0^1 xyf(x, y) dx dy$$

$$= \int_0^1 \int_0^1 xy(x + y - 2xy) dx dy$$

$$= \int_0^1 \int_0^1 (x^2y + y^2x - 2x^2y^2) dx dy$$

$$= \frac{2}{9}$$

Now

$$E(X)E(Y) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

And

$$E(XY) = \frac{2}{9}$$

Since $E(XY)$ is not equal to $E(X) \times E(Y)$, the variables are not independent.

Example: Given the following density function of X and Y:

$$f(x, y) = 4xy, \quad 0 < x < 1$$

= 0, elsewhere

Obtain $E(X)$ and $E(Y)$.

Solution:

$$E(X) = \int_0^1 \int_0^1 xf(x, y) dx dy = 4 \int_0^1 \int_0^1 x^2y dx dy = 2 \int_0^1 x^2 dx = \frac{2}{3}$$

$$E(Y) = \int_0^1 \int_0^1 yf(x, y) dx dy = 4 \int_0^1 \int_0^1 x^2y dx dy = \frac{4}{3} \int_0^1 x dx = \frac{2}{3}$$